DesignVis: A Novel Visual Interactive Design Tool Using Deep Learning Based on Visual Importance

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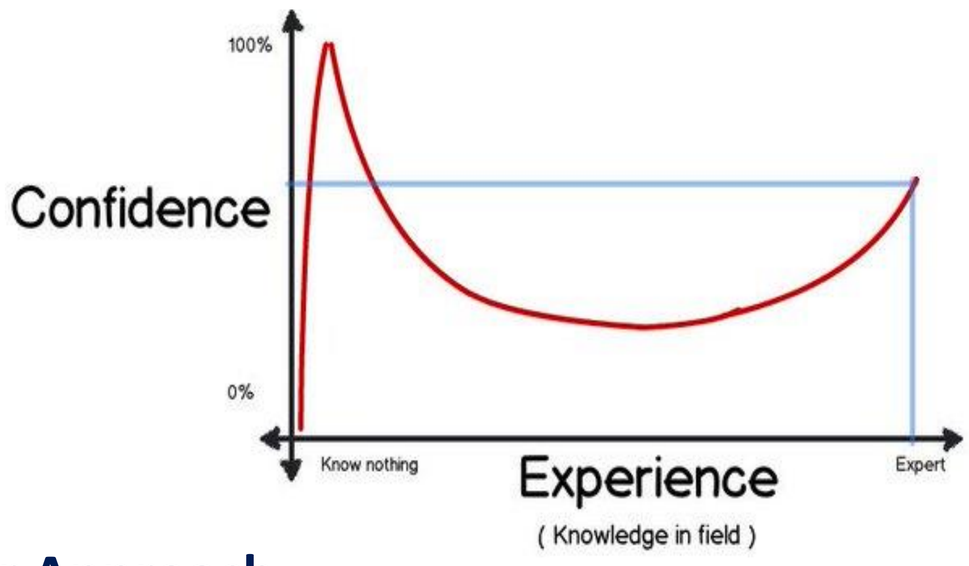
Introduction

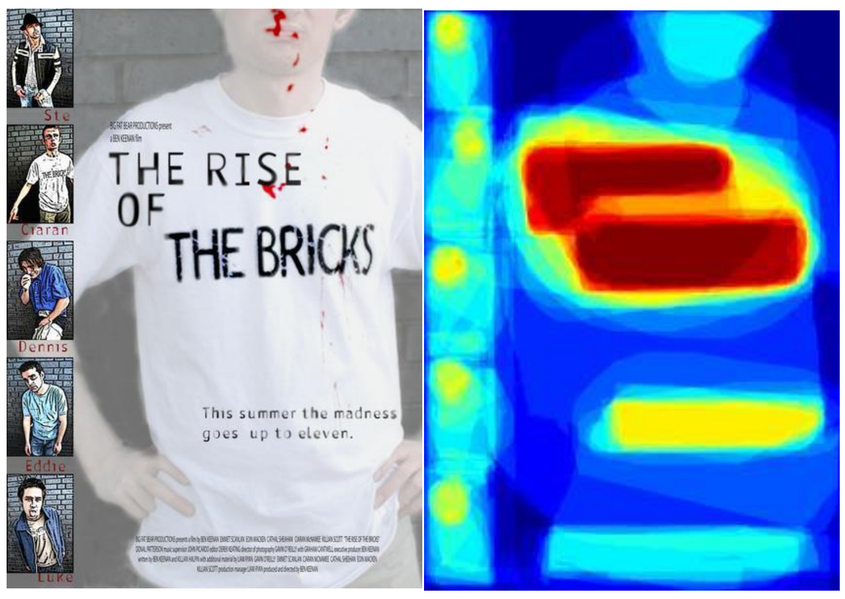
Design is indispensable for the life of modern people. But most of us are beginners in design. According to the Dunning-Kruger effect, people with poor design skills are overestimating their own works, and conversely, those with good skills are highly likely to underestimate the works.



Our Approach

Data

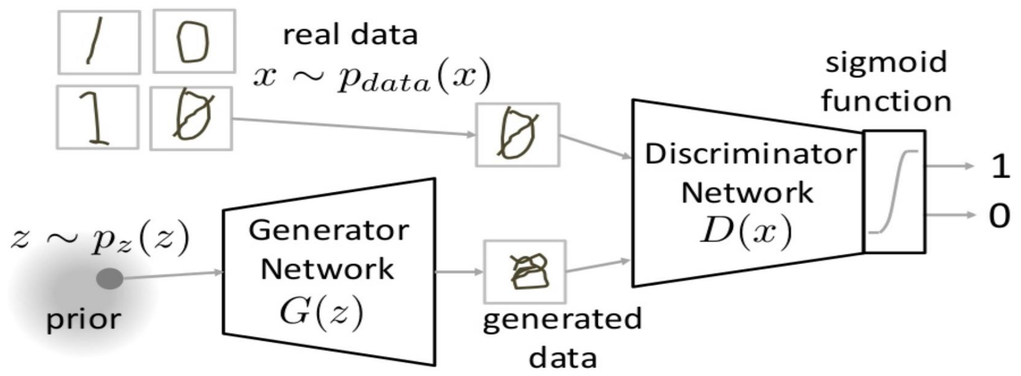
We adopted O'Donovan's GDI dataset [3] as training data. This dataset consists of 1,078 graphic designs collected from Flickr. These graphical designs are labeled, which is the area that 35 participants from Mechanical Turk regarded as important and labeled through a binary mask. At this time, the participants did not be inform with definition of the "important". The average value of 35 labels was adopted as a representative value.



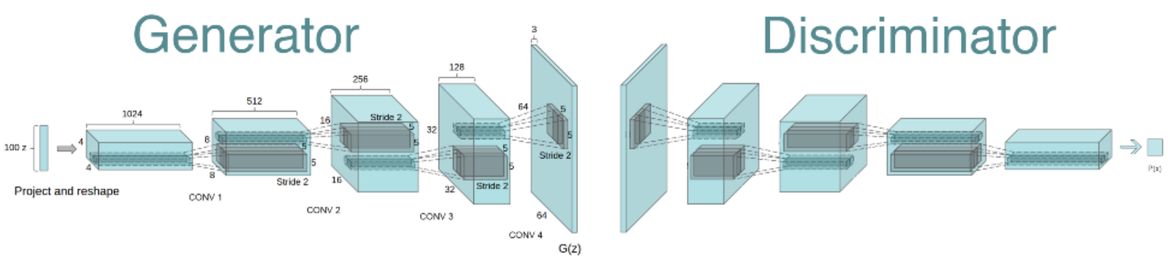
An example of a GDI dataset. Our goal is to learn GAN by using the single page graphic design data labeled with visual importance.

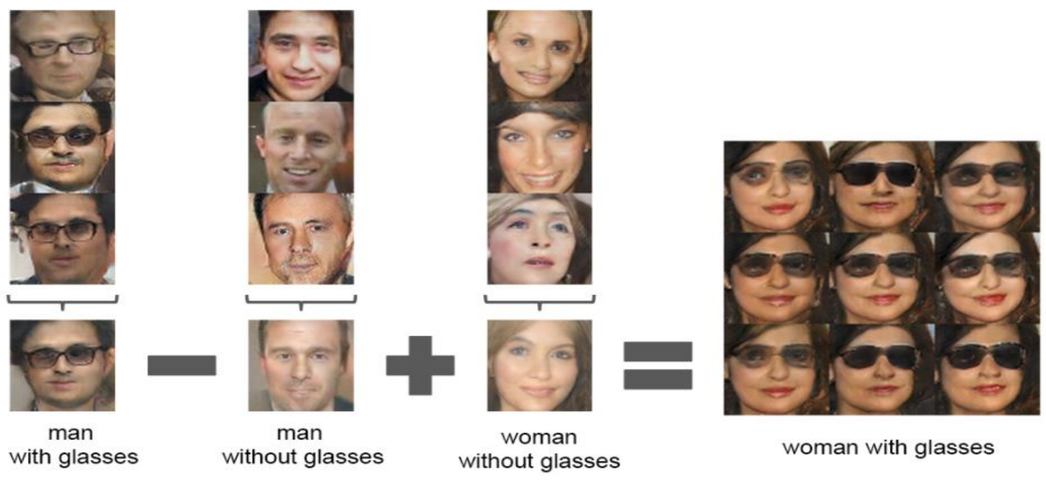
Model

$$V(D,G) = \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]$$



1.The generator network tries to generate data that is as close to real data as possible.
2.The discriminator network is trying to distinguish the two.

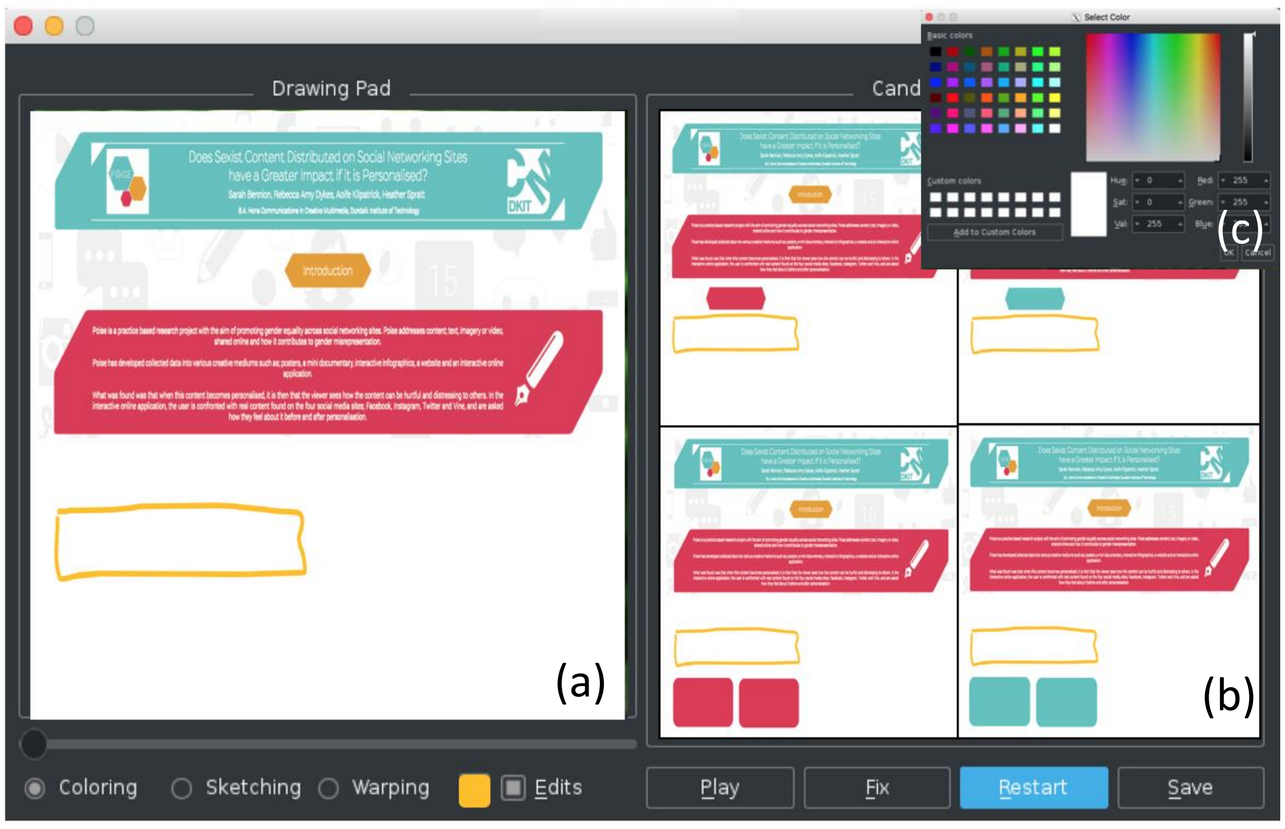




As we have seen, it is difficult to evaluate your own design, regardless of your design skills. Therefore, we propose a design tool named with DesignVis to support interactive design recommendation and evaluation.

In order to implement this, we studied Fully Convolutional Network and DiscoGAN model using GDI (Graphic Design Importance) data set. The former is needed to show Visual Importance, the latter to transform the user's rough sketch into design.

Implementation



(a): Space where users edit their design.
(b): GAN recommends elements to be added to the current design.
(c): A color palette for interactive color selection

The right figure show visual importance of the left. User can turn visual importance back on in our tool for evaluating their own design.

Future Work

- The DCGAN lacked capacity to learn the design data. We plan to develop a new GAN optimized for design data.
- We plan to reimplement the tool with web-based from QT-based.
- In order to evaluate DesignVis, we will do User study or Case study.

References

[1] iGAN : Zhu, Jun-Yan, et al. "Generative visual manipulation on the natural image manifold." *European Conference on Computer Vision*. Springer International Publishing, 2016.
[2] Visual Importance : Bylinskii, Zoya, et al. "Learning Visual Importance for Graphic Designs and Data Visualizations." *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*. ACM, 2017.
[3] GDI : O'Donovan, Peter, Aseem Agarwala, and Aaron Hertzmann. "Learning layouts for single-pagegraphic designs." *IEEE transactions on visualization and computer graphics* 20.8 (2014): 1200-1213.